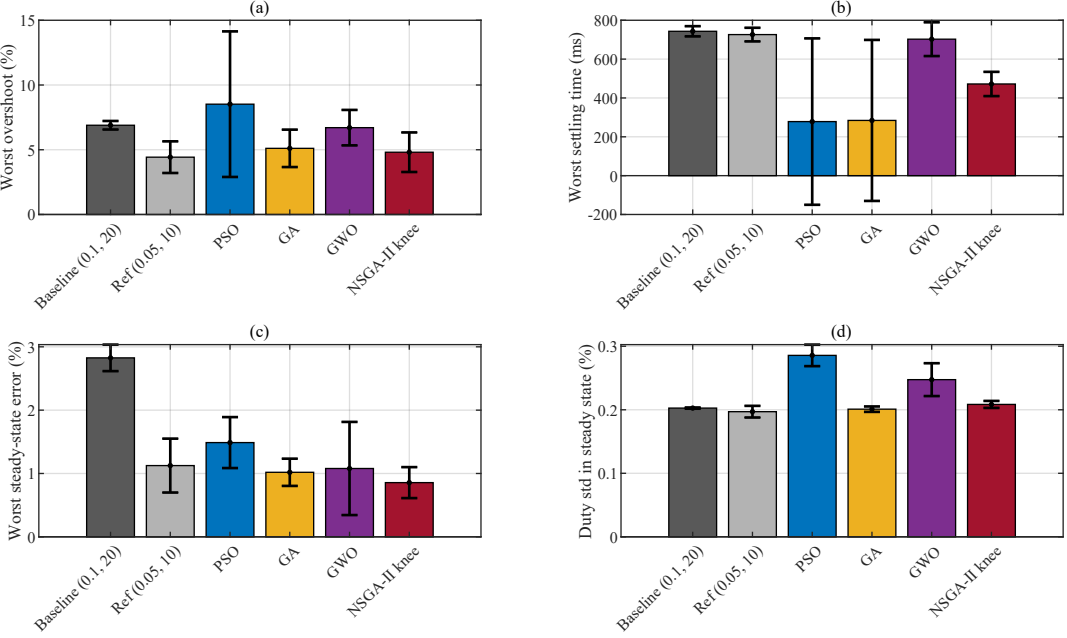

(a)
Worst overshoot (%)
15
10
5
0
(b)
Worst settling time (ms)
800
600
400
200
0
-200
(c)
Worst steady-state error (%)
3
2
1
0
(d)
Duty std in steady state (%)
0.3
0.2
0.1
0
Baseline (0.1, 20)
Ref (0.05, 10)
PSO
GA
GWO
NSGA-II knee